



HEALTH, SAFETY & ENVIRONMENT

Module 1: Introduction to Health and Safety

Definitions

Health and safety

Refers to all aspects of maintaining a healthy and safe workplace where harm to people is prevented

Safety

The results of ALL persons taking positive actions to identify accident causes and implement suitable preventative measures

Welfare

The availability of facilities and presence of conditions required for reasonably comfortable, healthy and secure living

Definitions

Accident, dangerous occurrence, near-miss and work-related ill-health

Accident

An unplanned occurrence which results in some loss, often an injury

Dangerous occurrence

“A readily identifiable event as defined under national law and regulations, with potential to cause injury or disease to persons at work or the public”

(International Labour Organisation)

Definations

Near miss

An incident in which no injury or damage results

Work-related ill-health

Any physical or psychological ill-health which is caused by or affected by your work

Size of the Problem

- According to the European Statistics on Accidents at Work (ESAW), every year in the 15 Member States of the European Union (EU) before the enlargements of 2004 and 2007 about 5,000 workers were killed in accidents at work and about 5 million workers were victims of accidents at work leading to more than three days' absence from work (EU, 2004).
- In India and China, the rates of occupational fatalities and accidents are similar at, respectively, 10.4 and 10.5 per 100,000 for fatalities, 8,700 and 8,028 for accidents.
- In sub-Saharan Africa, the fatality rate per 100,000 workers is 21 and the accident rate 16,000. This means that each year 54,000 workers die and 42 million work-related accidents take place that cause at least three days' absence from work.
- In Latin America and the Caribbean, about 30,000 fatalities occur each year and 22.6 million occupational accidents cause at least three days' absence from work.

Size of the Problem

Box 2 The burden and cost of occupational accidents and diseases in the United Kingdom

The Health and Safety Executive (HSE) has developed a methodology to calculate the costs of workplace accidents and diseases to individuals, employers and society. Each year:

- over 1 million injuries and 2.3 million cases of ill health occur;
- around 40 million working days are lost;
- over 25,000 individuals lose their jobs because of injury or ill health.

The estimated costs of this toll are:

- to individuals, US\$20–28 billion;
- to employers, US\$8–16 billion;
- to society, US\$40–60 billion.
- of a fatality: US\$3 million
- of a major injury: US\$80,000
- of an average case of ill health: US\$17,000

Source: Annual data based on 2005 statistics published by the HSE: <http://www.hse.gov.uk/costs/>; HSE, "Costs to Britain of workplace accidents and work-related ill health" (interim update), data based on 2000–2001 statistics: <http://www.hse.gov.uk/economics/costing.htm>.

Size of the Problem

Variations in performance affecting:

- Countries
- Economic Sector
- Size of Enterprise

Size of the Problem

Countries

- The incidence of workplace fatalities varies enormously between countries. There appears to be a significant difference between developed and developing countries:
 - a factory worker in Pakistan is eight times more likely to be killed at work than a factory worker in France;
 - fatalities among transport workers in Kenya are ten times those in Denmark;
 - construction workers in Guatemala are six times more likely to die at work than their counterparts in Switzerland (World Bank, 1995).

Size of the Problem

Economic sectors

- OSH performance varies significantly between economic sectors within countries. Statistical data show that, worldwide, the highest rates of occupational deaths occur in agriculture, forestry, mining and construction.
- The ILO has estimated, for example, that tropical logging accidents cause more than 300 deaths per 100,000 workers.
- In other words, three out of every 1,000 workers engaged in tropical logging die annually or, from a lifetime perspective, on average every tenth logger will die of a work-related accident.
- Similarly, certain occupations and sectors, such as meat packaging and mining, have high rates of work-related diseases, including fatal occupational diseases.

Size of the Problem

Sizes of enterprise

- Generally, small workplaces have a worse safety record than large ones. It seems that the rate of fatal and serious injuries in small workplaces (defined as those with fewer than 50 employees) is twice that in large workplaces (defined as those with more than 200 employees).

Reasons for Managing Health & Safety

- Moral
- Financial
- Legal

Reasons for Managing Health & Safety

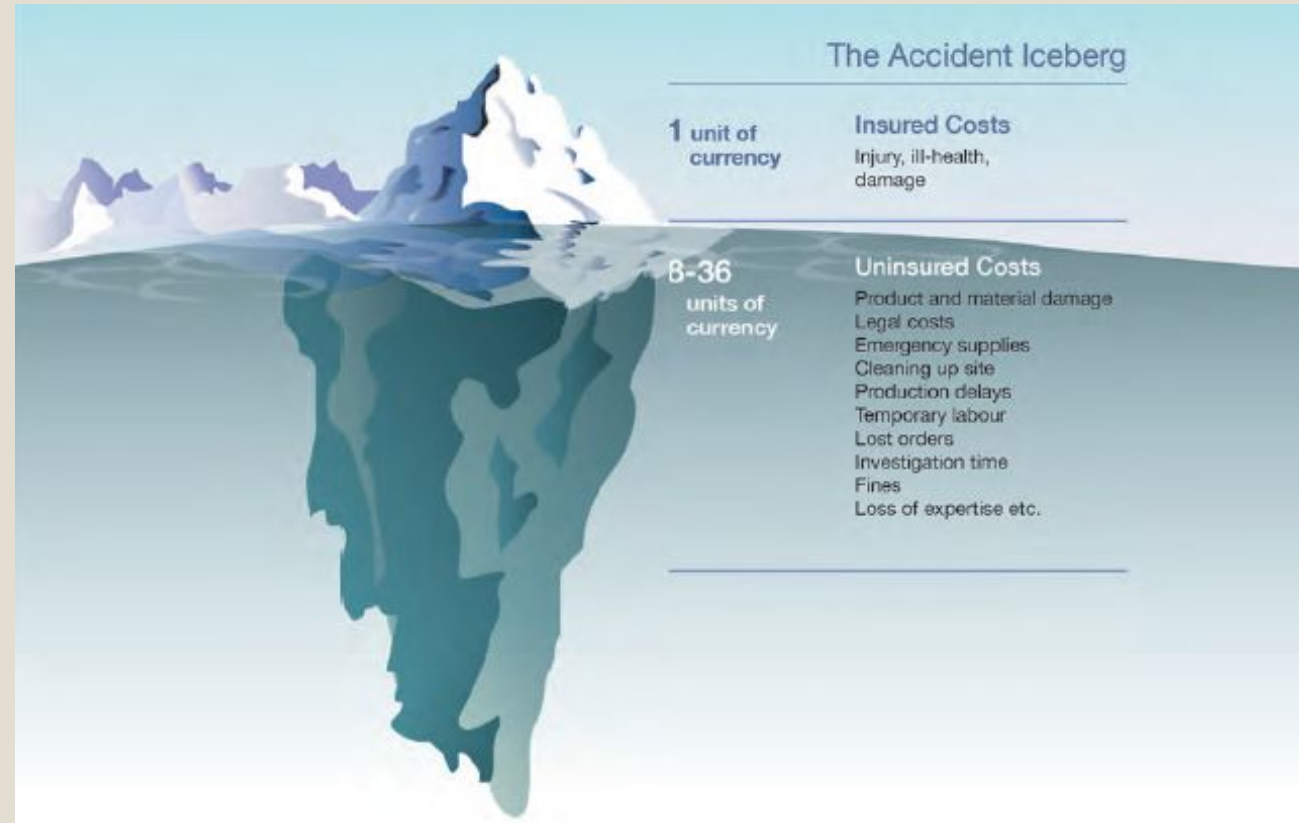
- Moral

Moral / ethical / humanitarian – trying to do the right thing

In 2005 the ILO estimated that as many as 2.2 million people die each year as a result of work-related accidents and diseases

Reasons for Managing Health & Safety

- Financial :
- Direct Cost
- Indirect Cost



Activity

What sort of financial costs would a company incur as a result of having poor health and safety standards?

The Buncefield oil storage depot incident

- Occurred on December 11th 2005
- Leak from a storage tank caused a large explosion
- 45 workers injured
- Costs of the incident estimated to be over £900 million
- Additional unquantifiable costs also significant

Financial costs

- Accident investigation costs
- Sick pay
- Lost time
- Damages and repairs
- Fines and legal costs
- Increased insurance premiums and compensation to victims

Non-financial costs

- Reputation with customers
- Reputation with neighbours
- Poor staff morale

Example

A worker has tripped over some boxes in a corridor because the light in that area was not working correctly

The worker suffered a broken arm and was off work for four weeks

The organisation knew about the problem with the light but had left it because they had decided that it was too expensive to call out an electrician who had quoted them £150 to complete the repair

The organisation was prosecuted in court by the enforcing agency and was convicted

Example

Action	Cost
Accident investigation costs by consultant	£500
Lost time and cost of agency worker to cover	£1184
Continued pay for the worker involved	£1600
Repair costs to light after accident	£150
Production loss cost	£1200
Increased insurance premium	£500
Fine for accident	£3000
Legal costs	£1000
Total Cost	£9134

Reasons for Managing Health & Safety

- Legal:
- Health and Safety at Work Act
- Role of ILO
 - Conventions
 - Recommendations

Who is responsible for health and safety?

Everyone

- But, this must be qualified
- How responsible?
- Responsible for what . . . exactly?

Health and safety responsibilities

Director / Senior Manager

- Develops policy and strategy
 - Provides resources and establishes targets
 - Develops the framework
 - Promotes a positive culture
 - Leads by example
 - Ensures the health, safety and welfare of staff

Health and safety responsibilities

Manager

- Establishes departmental standards and training needs
- Sets departmental and individual objectives
- Carries out periodic inspections
- Carries out risk assessments
- Manages issues and concerns
- Investigates accidents and cases of ill-health
- Ensures the health, safety and welfare of staff

Health and safety responsibilities

Supervisor

- Trains and instructs staff
- Carries out daily checks
- Ensures compliance with rules and procedures
- Assists with inspections and risk assessments
- Remedies defects
- Ensures the health, safety and welfare of direct reports

Health and safety responsibilities

Employee

- Complies with rules and procedures
- Co-operates with supervisor's instructions
- Maintains good housekeeping standards
- Looks after health and safety of self and others
- Reports unsafe practices and conditions
- Makes suggestions for improvement
- Maintains an active interest in health and safety initiatives

Health and safety responsibilities

Competent Assistance

- Provides assistance and advice
- Can be external or part-time
- No legal responsibility

Competency

- Sufficient knowledge, experience, training and qualifications
- Other qualities

Health and safety responsibilities

Some of the functions performed by the competent assistance are:

- Developing policies, procedures and systems of work
- Advising on legal compliance
- Planning training initiatives
- Carrying out training
- Reviewing performance
- Assisting with risk assessments
- Liaising with external bodies

Communicating the safety message

Staff member

- Informal / verbal (grapevine, chatting)
- Occasional written (incident report form)

Supervisor / Manager

- More formal / written
- Inspection reports, performance reports
- Presentations and meetings

Remaining alert and dealing with issues

As a manager / supervisor you should:

- Remain vigilant
- Understand the hazards and risks in the workplace
- Be prepared to correct behaviours of others as necessary
- Recognise and praise good behaviours
- Respond to concerns
- Take immediate action where necessary
- Lead by example

Risk Management

- To ensure a safe work environment, you need to understand the definition of **risk**.
- To define risk, we first need to take a look at a few basic terms...
- A **hazard** is anything that has the potential to cause **harm**
- For a hazard to cause harm, a **hazardous event** must happen

Risk Management

- The **likelihood** is the chance that the hazardous event will occur
- **Consequence** is the outcome of the hazardous event
- **Risk** is the combination of the **likelihood** of a hazardous event occurring, and the **consequence** of the event
- **Risk = Likelihood x Consequence**

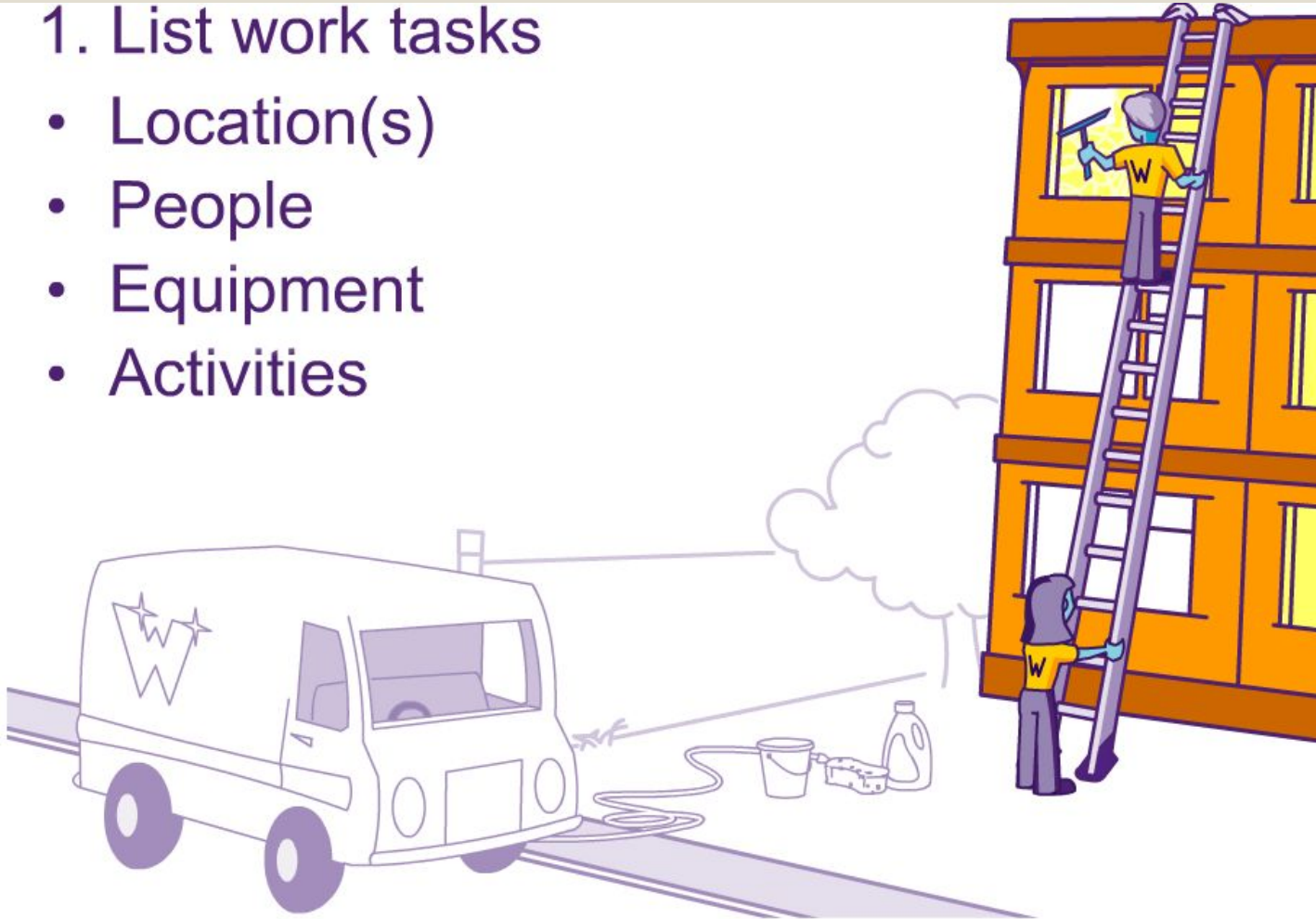
Risk Assessment

- A risk assessment is a careful examination of anything in your work environment that could cause injury or ill health

Risk Assessment

1. List work tasks

- Location(s)
- People
- Equipment
- Activities



Risk Assessment

2. Identify the risks

- What are the hazards?
- Who might be harmed?
- How might they be harmed?

Risk Assessment

3. Estimate the risk

- You need to consider two things:
- how likely is it that something could go wrong?
- how serious would the outcome be?

Likelihood

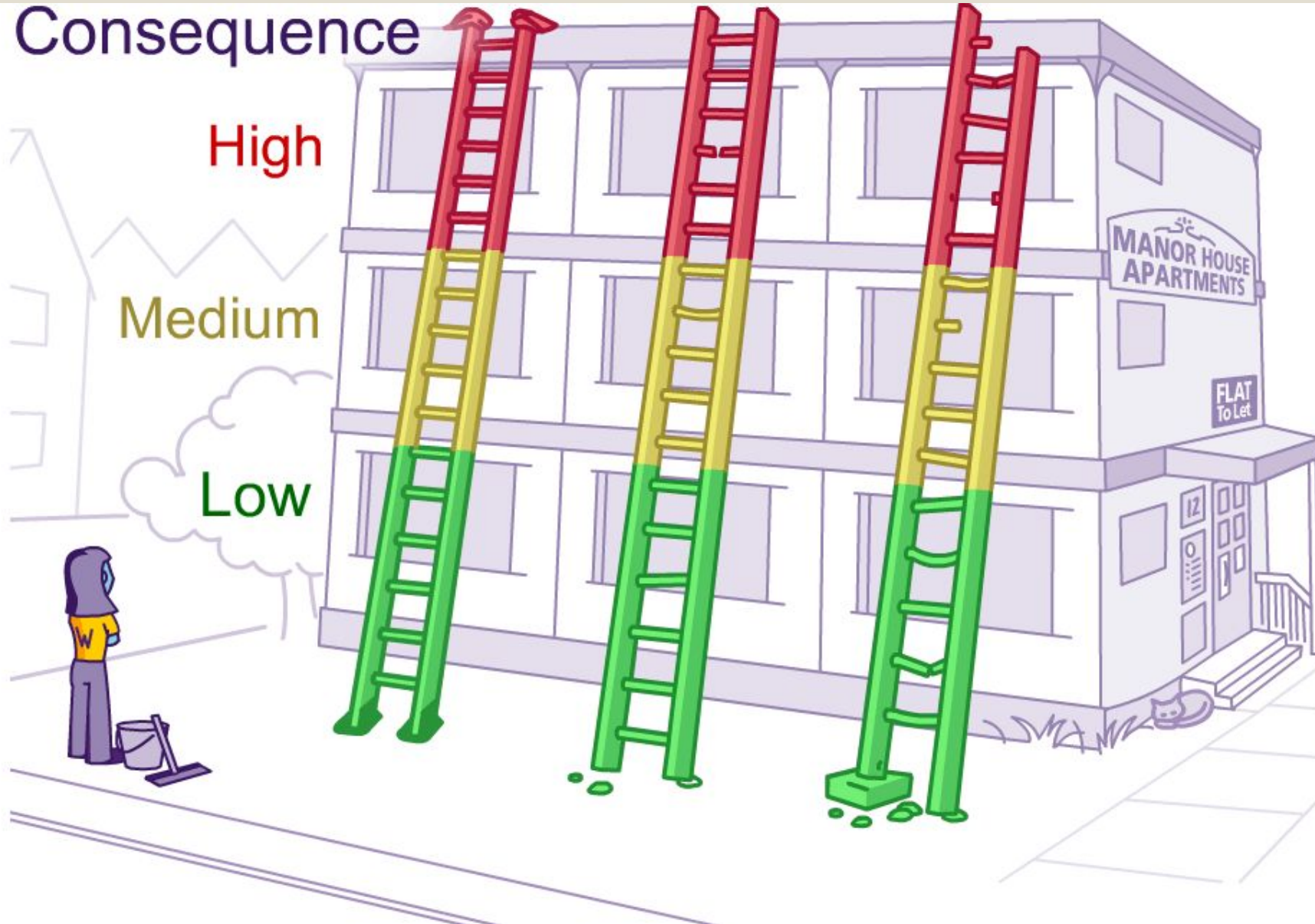


Consequence

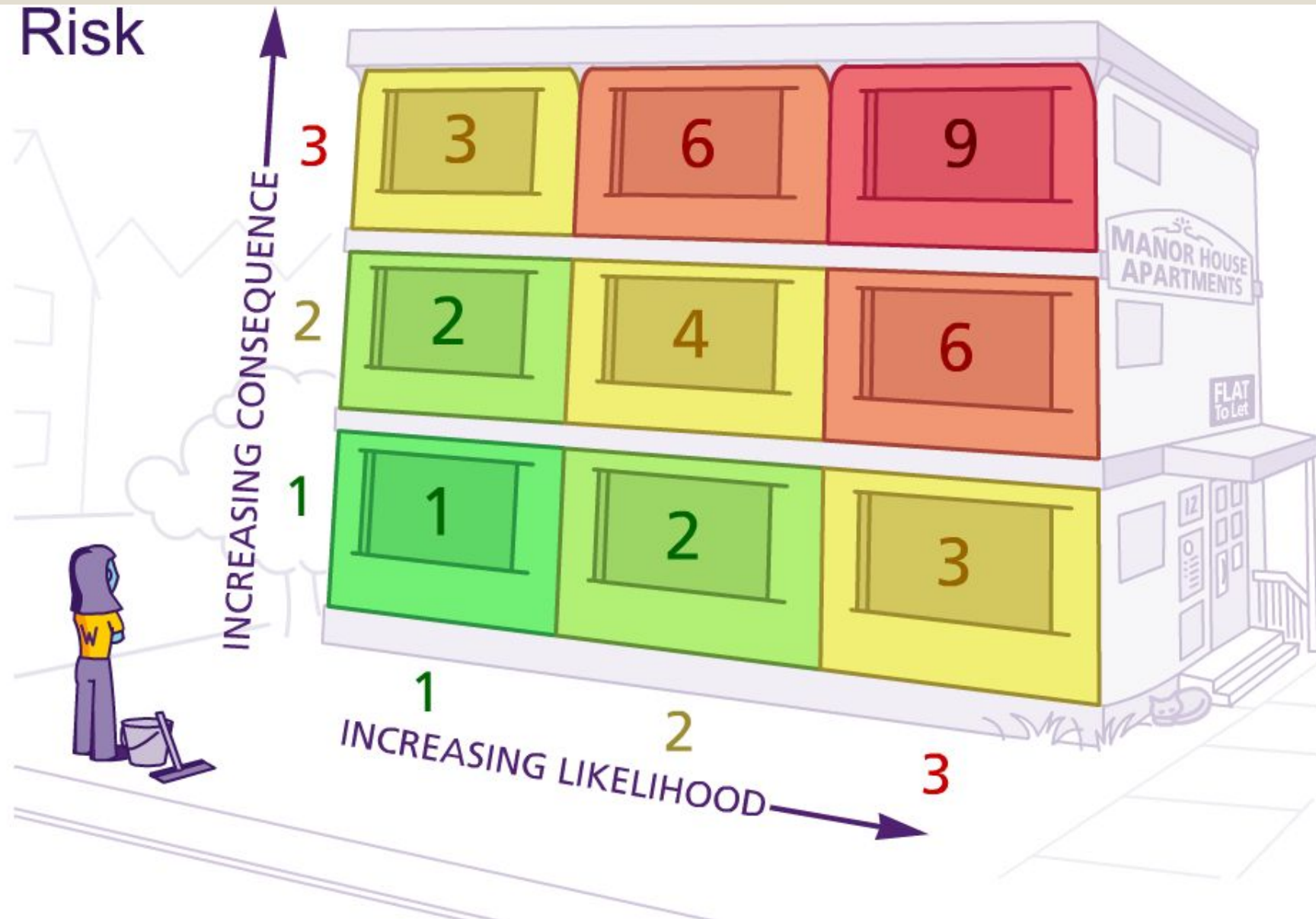
High

Medium

Low

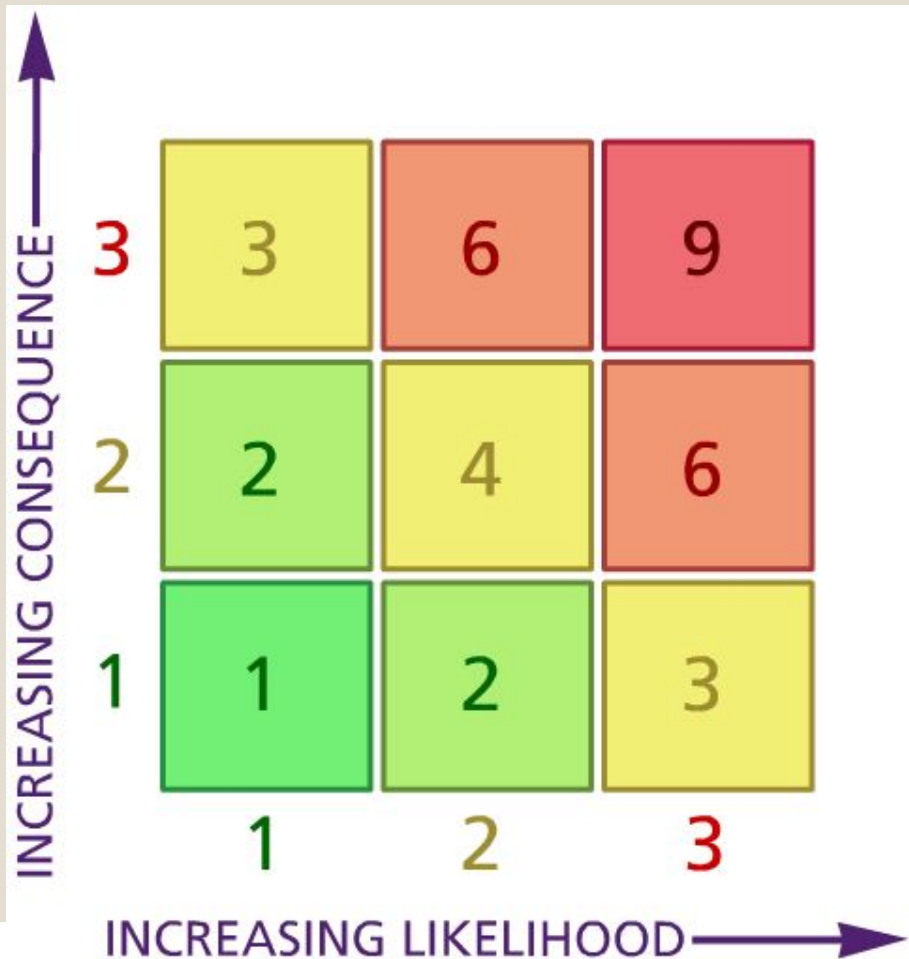


Risk



Risk Assessment

4. Evaluate the risk



6-9 Unacceptable

Take immediate action

3-4 Tolerable

Look to improve

1-2 Acceptable

No further action, but
ensure controls are
maintained

Risk Assessment

5. Record your findings

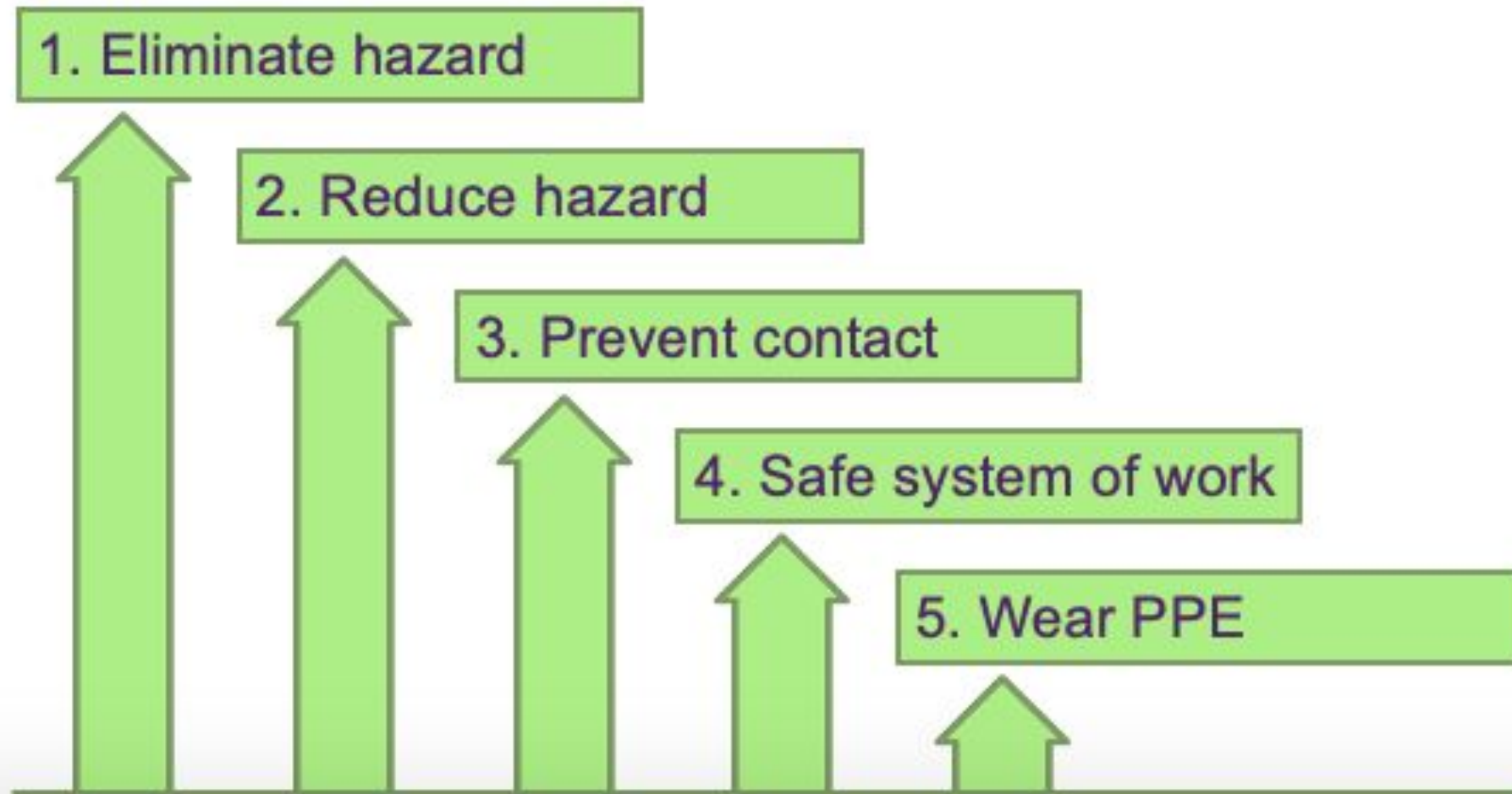
- the location, activity and equipment being assessed
- hazards and risk levels
- risk controls
- assessor's details
- date and time
- review date

Risk Assessment

6. Review your findings

- When changes occur (personnel, equipment or activities), you should review your risk assessments.
- It's a legal requirement to review assessments regularly

To help decide on the risk control there is a **hierarchy of risk control** that you can use



Principles to prevent Accident

The Ministry of Labour and National Service¹ postulated six principles of accident prevention in 1956 that are still valid today. These are:

- 1 Accident prevention is an essential part of good management and of good workmanship.
- 2 Management and workers must co-operate wholeheartedly in securing freedom from accidents.
- 3 Top management must take the lead in organising safety in the works.
- 4 There must be a definite and known safety policy in each workplace.
- 5 The organisation and resources necessary to carry out the policy must exist.
- 6 The best available knowledge and methods must be applied.

Accident Causation Models

- Heinrich states that the occurrence of an injury accident invariably results from a completed sequence of factors culminating in the accident itself. He postulates five factors or stages in the accident sequence.

The five factors or stages in the sequence of events are:

- (a) ancestry and social environment, leading to
- (b) fault of person, constituting the proximate reason for
- (c) an unsafe act and/or mechanical hazard, which results in
- (d) the accident, which leads to
- (e) the injury.

Accident Causation Models

- Bird and Loftus have extended this theory to reflect the influence of management in the cause and effect of all accidents that result in a wastage of the company's assets. The modified sequence of events becomes:
- (a) lack of control by management, permitting
- (b) basic causes (personal and job factors), that lead to
- (c) immediate causes (substandard practices/conditions/errors), which
- are the proximate causes of
- (d) the accident, which results in
- (e) the loss (minor, serious or catastrophic).

Multiple Causation Theory

